UGANDA CHRISTIAN UNIVERSITY

The University of Africa

Department of [TO CONFIRM]

Machine Learning for IoT-Based Soil Health Prediction

PimaUdongo · Robustness of reported skill under spatial hold-out

Adrian Rugaba · [REG NO: TO CONFIRM]

Supervisor: [TO CONFIRM]

BACKGROUND

Soil-carbon measurement is slow and expensive. Laboratory analysis of total carbon from five soil properties is the standard, but it cannot keep pace with the number of sites IoT soil-health monitoring now demands.

PROBLEM

A reported score does not say whether the model would still work if whole regions were held out. Random splitting mixes nearby sites into training and test, so the reported skill can be optimistic.

RESEARCH GAP

Evaluations rarely combine a geographically blocked hold-out with an uncertainty statement about the prediction. Both are needed before a model can be trusted on a sensor network.

AIM

Measure how much of the reported skill depends on spatial structure, and whether the model can say when it does not have enough evidence.

RESEARCH QUESTION

For total carbon predicted from five laboratory soil properties, how much accuracy is lost when whole areas are held out, and can the model flag low-confidence predictions?

SCOPE

926 samples · 58 sites · 16 clusters · 19 countries. Random forest, five seeds. Spatial hold-out by cluster, plus split-conformal intervals.

RESEARCH OBJECTIVES

- Quantify accuracy loss under a spatial hold-out.
- Compare random and cluster-blocked evaluation.
- Calibrate prediction intervals on held-out data.
- Test tolerance to simulated sensor error.
- State the limits of the evidence clearly.

METHODOLOGY

1. **Assemble**

   926 soil samples from 58 sites across 16 clusters and 19 countries.

2. **Split**

   Random split for the baseline; whole clusters held out for the spatial test.

3. **Train**

   Random forest, five seeds, tuned on the training folds only.

4. **Evaluate**

   R^2 and RMSE under both splits; RMSE against the field mean.

5. **Calibrate**

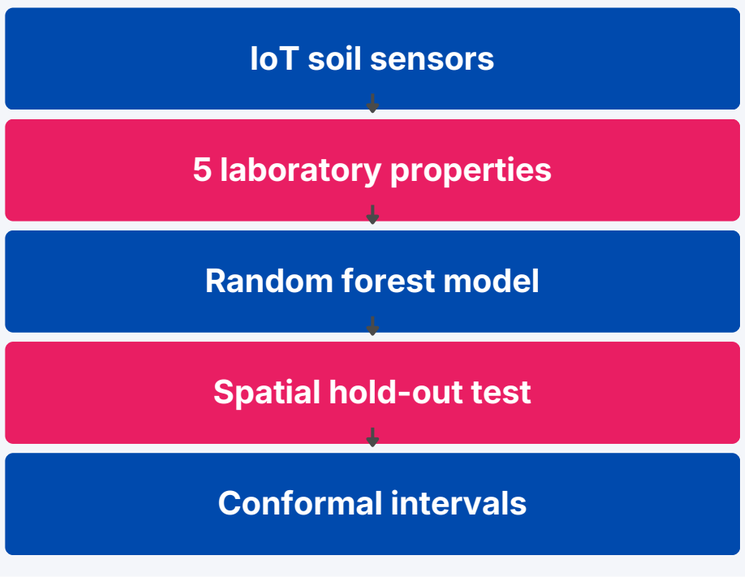
   Split-conformal intervals on a held-out calibration set.

KEY FINDING

0.252

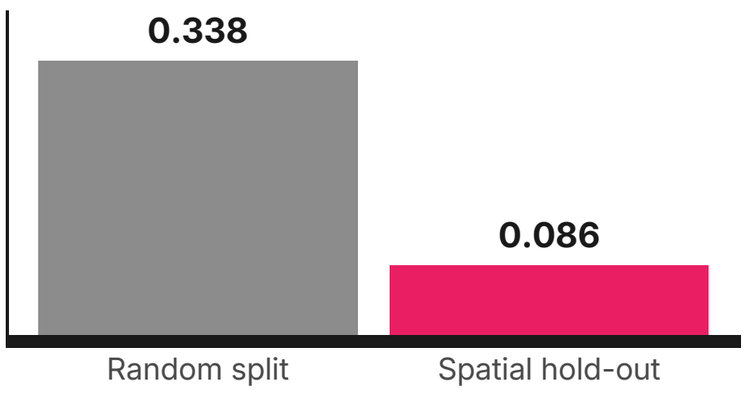
drop in R^2 when whole clusters are held out: 0.338 random → 0.086 spatial.

SYSTEM PIPELINE



MAIN RESULTS

R^2 by evaluation split



Spatial RMSE 1.339 against 1.421 for predicting the field mean — the model still beats the naive baseline, but by less than half the margin the random split suggested.

ERROR TOLERANCE

Adding noise at 0.25 of the training standard deviation leaves R^2 at 0.039; at 0.50 it falls to −0.095. The model is fragile to sensor error.

LIMITATIONS

Coverage of 88.8% against a 90% target, with a mean interval width of 4.21. Not validated on field sensors or Ugandan farms.

CONCLUSION

Reported accuracy is strongly spatially dependent. Before IoT soil-health prediction can be deployed, evaluation must hold out whole regions and state uncertainty explicitly.